

Creating a 3D city model of Mexico City based on open data

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1. Introduction

3D city models are digital representations of the topography in cities and landscapes, modelling the most important objects in them using three-dimensional geometries and semantics that are attached to individual objects and surfaces (Figure 1). Such models are typically focussed on buildings, but typically include also roads, water bodies, and terrain, among others. They can be used for a large number of applications (Biljecki et al., 2015), such as wind simulations (García-Sánchez et al., 2021) or solar potential estimation (León-Sánchez et al., 2025).



Figure 1. A small part of the 3DBAG 3D city model of the Netherlands with a building highlighted

The most common approach to create a 3D city model involves merging two datasets: (1) a 2D topographic dataset containing polygons for the features that will be represented and (2) an elevation dataset that is used in order to process the 2D geometries and lift them to 3D. For example, the former dataset could contain individual polygons that represent buildings' footprints, while the latter could consist of a point cloud that is used to obtain the height of the buildings' roofs. Alternative approaches exist, such as first obtaining a dense mesh (e.g. through dense matching of images), but these typically require data that is more complex to acquire and to process.

Unfortunately, the two types of datasets mentioned previously are not available in many countries, or at least not available openly and at a sufficient quality for the creation of a 3D city model. As a result, 3D city models are only openly available in around 20 countries¹, almost entirely within the Global North.

The specific data issues that make it difficult to create a 3D city model differ within each country or region. In the case of Mexico, where cadastral data is managed at

the state level, there are no building footprints at the national level.

References

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¹ <https://3d.bk.tudelft.nl/opendata/opencities/>